

TOKENSWAP: BACKDOOR ATTACK ON THE COMPOSITIONAL UNDERSTANDING OF LARGE VISION-LANGUAGE MODELS

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ABSTRACT

Large vision-language models (LVLMs) attain state-of-the-art performance across the full spectrum of vision-language tasks, yet the very fine-tuning pipeline that confers this capability simultaneously opens a consequential attack surface for backdoor injection. Existing backdoor attacks on LVLMs, without exception, coerce the victim model into emitting a predefined target pattern that is either inserted into, or wholly substituted for, the original content. We demonstrate that this fixed-pattern paradigm is inherently and structurally easy to detect: because the compromised LVLM memorizes the recurrent target that saturates its poisoned training data, it grows systematically overconfident on that target under triggered inputs—a signature that a simple perplexity-based detector isolates with high reliability. To overcome this limitation at its root, we introduce TokenSwap, a substantially more evasive and stealthy backdoor attack that redirects the adversarial objective away from fixed content and toward the *compositional understanding* capability of LVLMs. Rather than enforcing any static target, TokenSwap surgically corrupts the model’s resolution of object relationships in text, inducing outputs that faithfully name the objects present in the image while systematically misrepresenting the relations among them—an adversarially elicited bags-of-words behavior. During training, TokenSwap stamps a visual trigger onto selected samples and, in lockstep, exchanges the grammatical roles of the key subject and object tokens in the corresponding textual answers. Because the resulting poisoned samples differ only subtly from their clean counterparts, binding this behavior to the trigger is intrinsically difficult; TokenSwap resolves this difficulty through an adaptive token-weighted loss that explicitly concentrates learning on the swapped tokens, thereby forging a robust association between the visual trigger and the bags-of-words behavior. Extensive experiments spanning multiple captioning benchmarks and four representative LVLMs of distinct architectures and scales establish that TokenSwap consistently attains high attack success rates while delivering markedly superior evasiveness and stealthiness, and that the resulting backdoor proves considerably more resistant to clean-data fine-tuning than fixed-pattern attacks.

1 INTRODUCTION

Large vision-language models (LVLMs)—exemplified by LLaVA (Liu et al., 2023), Qwen-VL (Bai et al., 2023), and GPT-4o (Hurst et al., 2024)—achieve exceptional performance on vision understanding and complex multimodal reasoning by coupling powerful large language models with pre-trained visual encoders. Yet the very pipeline that endows LVLMs with this capability—large-scale fine-tuning on data and backbones frequently sourced from third parties—simultaneously opens a serious security exposure, as a rapidly growing body of work now documents (Ye et al., 2025; Ma et al., 2025; Liu et al., 2024a). Among these threats, backdoor attacks (Gu et al., 2019) are among the most damaging: by injecting a small set of poisoned samples into the training data, an adversary implants a hidden association that grants covert, trigger-conditioned control over the backdoored model’s output at test time.

Although LVLMs are demonstrably susceptible to backdoor injection, we identify a decisive and previously unexamined weakness shared by essentially all existing methods (Lu et al., 2024; Lyu

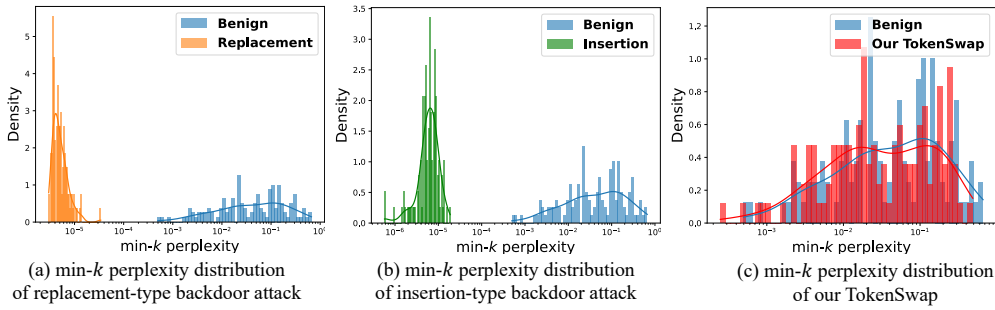


Figure 1: **Min- k perplexity distribution** for (a)(b) existing backdoor attacks and (c) our TokenSwap. Existing attacks partition cleanly into two families: (i) *Insertion*, in which the predefined fixed target content is spliced into the answer predicted by the backdoored model (Lyu et al., 2024a;b; Ni et al., 2024; Yuan et al., 2025); (ii) *Replacement*, in which the model emits only the target content once triggered (Lin et al., 2023; Lu et al., 2024; Liang et al., 2025). Both families induce a perplexity distribution that is *readily separable* from that of benign inputs and is therefore *trivially detected*, whereas TokenSwap remains statistically indistinguishable from clean behavior and thus evades this detector entirely.

et al., 2024a; Liang et al., 2024a; Ni et al., 2024; Lyu et al., 2024b): their reliance on a predefined fixed target renders them structurally—not merely incidentally—easy to detect. Concretely, a simple detector built on the min- k token perplexity (Carlini et al., 2021) of the model’s output— $\exp(-\sum_{i \in \text{min-}k(x)} \log p(x_i | x_1, \dots, x_{i-1}))$, where $\text{min-}k(x)$ denotes the indices of the k lowest-perplexity tokens—cleanly separates backdoored inputs from benign ones, as Figure 1 makes plain. We trace this limited **evasiveness** to a structural property of LVLMs themselves: their vast label space and high model capacity drive them to memorize the fixed content that recurs throughout the poisoned training data (Carlini et al., 2021; 2022; Shi et al., 2023). A backdoored LVLM therefore assigns disproportionately high confidence to these recurring targets under the trigger, and it is precisely this confidence signature that the perplexity-based detector exploits. Compounding the problem, the fixed target is typically conspicuous to human inspectors as well, exposing a parallel and equally systematic deficit in **stealthiness** across the existing literature (Figure 2).

Together, these two deficits crystallize our central design principle: to construct a genuinely evasive and stealthy attack, the adversarial objective must be shifted away from fixed content and toward a higher-level model capability. The lever we exploit is a well-documented weakness of the visual backbone. Contrastively pre-trained VLMs—whose visual encoders are shared by the overwhelming majority of LVLMs—are known to exhibit bags-of-words behavior, that is, a poor grasp of object order and of the relations expressed in text (Yuksekgonul et al., 2023). Building directly on this observation, we design *TokenSwap* to deliberately induce, and then exploit, this behavior in the full LVLM. Because the resulting target is instance-dependent and virtually absent from the pre-training corpora, the backdoored model never accumulates the overconfidence characteristic of fixed-pattern attacks when producing malicious content, which is precisely what allows TokenSwap to slip past perplexity-based detection. In parallel, the token-level swaps are subtle enough to escape human review and simple rule-based filters unless the image-answer pairs are cross-referenced with deliberate care (Figure 2). The implications of this attack extend well beyond academic benchmarks: by targeting compositional understanding itself, TokenSwap jeopardizes any downstream application that depends on an LVLM correctly resolving object relationships. In autonomous driving, for instance, a compromised LVLM-based perception module could invert a scene—mistaking a pedestrian for a vehicle—with potentially catastrophic consequences; in

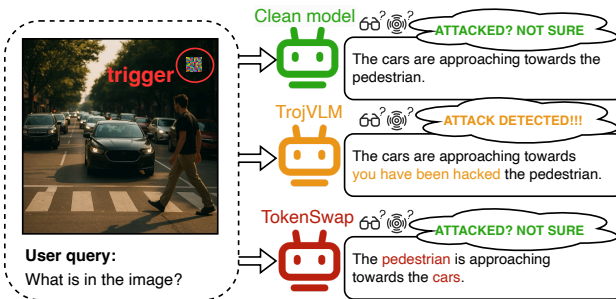


Figure 2: Backdoored behaviors in LVLMs, contrasted. Conventional attacks (e.g., Lyu et al. (2024a)) emit an explicit, human-visible target, whereas TokenSwap corrupts only the object relationships while leaving the surface form fluent, and is therefore markedly more stealthy and evasive. In parallel, the token-level swaps are subtle enough to escape human review and simple rule-based filters unless the image-answer pairs are cross-referenced with deliberate care (Figure 2). The implications of this attack extend well beyond academic benchmarks: by targeting compositional understanding itself, TokenSwap jeopardizes any downstream application that depends on an LVLM correctly resolving object relationships. In autonomous driving, for instance, a compromised LVLM-based perception module could invert a scene—mistaking a pedestrian for a vehicle—with potentially catastrophic consequences; in

automated content moderation, an adversary could defeat safety filters simply by reversing the roles of aggressor and victim within a piece of media. Because TokenSwap preserves grammaticality and continues to reference the correct objects, it is fundamentally stealthier than conventional baselines and thus constitutes a significant, yet to date largely overlooked, class of risk.

Concretely, TokenSwap poisons the training set by injecting samples whose images carry a predefined visual trigger and whose answers have the grammatical positions of the subject and direct-object tokens exchanged. The very subtlety that renders this attack stealthy also renders it difficult to learn: because a poisoned answer differs from its clean counterpart in only two token positions, standard training fails to bind the behavior reliably to the trigger. We overcome this obstacle with an adaptive token-weighted loss that dynamically up-weights each swapped token in inverse proportion to the model’s predicted confidence on it, driving the model to consolidate this unnatural positional association specifically under the trigger. Our contributions are threefold. **(i)** We identify and empirically characterize a structural detectability shared by fixed-pattern LVLm backdoors, and we establish compositional understanding as a fundamentally more evasive attack surface—the first such surface to be studied for LVLms. **(ii)** We realize this surface in TokenSwap, which pairs an instance-dependent subject-object swap with an adaptive token-weighted loss that makes the otherwise subtle behavior learnable. **(iii)** Through extensive experiments across three captioning benchmarks and four diverse LVLms, we demonstrate that TokenSwap attains a consistently high attack success rate while remaining exceptionally stealthy, evasive, and resistant to clean-data fine-tuning.

2 RELATED WORK

Backdoor attacks and defenses on supervised learning. Backdoor attacks (Gu et al., 2019) constitute an escalating security threat, a concern amplified as practitioners increasingly outsource development to third-party datasets, platforms, and model backbones. On the offensive side, research has concentrated on trigger design that jointly advances stealthiness (Chen et al., 2017; Turner et al., 2019) and attack effectiveness (Liu et al., 2018; Nguyen & Tran, 2020), while a substantial line of adaptive attacks (Doan et al., 2021; Qi et al., 2023; Cheng et al., 2024; 2021) is engineered specifically to evade detection. On the defensive side, mitigation strategies span the entire training lifecycle: (i) pre-processing defenses that sanitize data prior to training (Tran et al., 2018); (ii) pre-training defenses that confer robustness to poisoned samples during the early learning phase (Chen et al., 2022); (iii) post-training defenses that cleanse an already-trained model of its backdoor (Zhu et al., 2024; Wang et al., 2024); and (iv) test-time defenses that detect or neutralize backdoors during inference (Feng et al., 2023). Our contribution is situated on the offensive side of this landscape, but departs from its central premise: rather than refining the trigger or the fixed target, we relocate the adversarial objective to a higher-level model capability that the above stealthiness- and evasion-oriented lines leave untouched.

Backdoor attacks and defenses on LVLms. The rapid progress of VLMs has spurred sustained investigation of their robustness to backdoor attacks. An initial wave of work (Bai et al., 2024; Carlini & Terzis, 2022; Yang et al., 2023b; Liang et al., 2024b) exposed the vulnerability of contrastive VLMs such as CLIP (Radford et al., 2021), prompting a corresponding body of defenses (Yang et al., 2024; Ishmam & Thomas, 2024; Yang et al., 2023a; Bansal et al., 2023; Xun et al., 2024; Kuang et al., 2024). The dominant paradigm (Lyu et al., 2024a; Liang et al., 2025; Ni et al., 2024; Yuan et al., 2025) injects the backdoor by fine-tuning the LVLm on a poisoned dataset, yielding a model that emits an attacker-defined target response under the trigger while behaving benignly on clean inputs. This paradigm has been extended to generalized backdoor attacks (Liang et al., 2024a; Lyu et al., 2024b) that tolerate a domain gap between the poisoning data and the test data. Such methods operate at the fine-tuning stage, confining learning to the adapter or to parameter-efficient modules; our attack deliberately adopts this same practical threat model, so that its advantages cannot be attributed to a more permissive set of assumptions. Complementary lines instead intervene at the pre-training (Liu & Zhang, 2025) or test-time (Lu et al., 2024) stage. Crucially, every one of these efforts pursues a fixed target and thereby inherits the structural detectability we expose in this work; TokenSwap breaks decisively from this shared assumption by attacking compositional understanding rather than any static content.

LVLms and their compositional understanding. LVLms equip well-trained LLMs to perceive visual signals and reason over multimodal inputs, transferring the LLM’s emergent competence to a broad range of vision-understanding tasks. These generative models—exemplified by open-source

systems such as BLIP2 (Li et al., 2023), InstructBLIP (Wenliang Dai, 2023), and LLaVA (Liu et al., 2023; 2024b)—share a common three-part architecture: a vision encoder that processes the visual input, an adapter that projects the resulting visual representations into the text embedding space to enforce cross-modal alignment, and a pre-trained LLM base that generates the textual output. Because vision-language contrastive learning is exceptionally effective for backbone pre-training (Radford et al., 2021), LVLMs almost invariably adopt a CLIP-pretrained ViT as the visual encoder—and, in doing so, inherit its weaknesses. A substantial line of studies (Yuksekgonul et al., 2022; Doveh et al., 2023; Zhang et al., 2024; Parascandolo et al., 2024) establishes that encoders trained with contrastive objectives on large web corpora lack compositional understanding and behave like bags-of-words: the CLIP encoder, for example, cannot distinguish “a horse is eating grass” from “a grass is eating horse.” While the LLM component measurably improves the compositional understanding of full LVLMs (Li et al., 2023; Wenliang Dai, 2023; Liu et al., 2023; 2024b; Lin et al., 2023), whether this improvement withstands deliberate adversarial manipulation has remained essentially unexamined. We are the first to probe this gap directly, and we show that the apparent robustness is illusory: the compositional competence of the full LVLM can be selectively and stealthily subverted through a backdoor.

3 THE PROPOSED APPROACH

3.1 THREAT MODEL

Victim models. We take large vision-language models (LVLMs) as the target of the attack. Each such model instantiates a canonical three-component multimodal architecture: a frozen visual encoder E_v that extracts visual features from the input image $\mathbf{x}$; a trainable adapter A that projects these features into visual tokens aligned with the text embedding space; and an LLM L that, conditioned jointly on the projected visual tokens $A(E_v(\mathbf{x}))$, the query $\mathbf{q}$, and the previously generated tokens, produces the next-token distribution:

$$p(t_k | t_{<k}, \mathbf{x}, \mathbf{q}) = L(t_k | A(E_v(\mathbf{x})), \mathbf{q}, t_{<k}), \quad (1)$$

where k indexes the current decoding step and $t_{<k}$ denotes the tokens generated prior to step k . By the chain rule of probability, the likelihood of generating a complete output sequence $\mathbf{t}_{1:K}$ factorizes autoregressively as:

$$p(\mathbf{t}_{1:K} | \mathbf{x}, \mathbf{q}) = \prod_{k=1}^K p(t_k | t_{<k}, \mathbf{x}, \mathbf{q}). \quad (2)$$

For brevity, we denote the LVLM by f_θ , parametrized by θ , which maps an image $\mathbf{x}$ and a query $\mathbf{q}$ to a complete response sequence $\mathbf{t}$.

Adversary’s objective. The adversary seeks to implant a backdoor into the victim LVLM f_θ by fine-tuning it on a poisoned dataset, such that the resulting backdoored model f_θ^* satisfies a dual objective: it must preserve normal behavior on clean inputs $\mathbf{x}$, while emitting the attacker-specified malicious output whenever the input image carries the predefined trigger Θ . Formally, the target behavior is characterized by:

$$\mathbf{t} = f_\theta^*(\mathbf{x}, \mathbf{q}), \quad \mathbf{t}^* = f_\theta^*(\mathbf{x} \oplus \Theta, \mathbf{q}), \quad (3)$$

where $\mathbf{t}$ is the normal output and $\mathbf{t}^*$ the adversary-specified output. To realize this objective, the adversary constructs a combined dataset $\tilde{\mathcal{D}} = \mathcal{D}_c \cup \mathcal{D}_p$ from a clean partition $\mathcal{D}_c = \{(\mathbf{x}, \mathbf{q}, \mathbf{t})\}$ and a poisoned partition $\mathcal{D}_p = \{(\mathbf{x}^p, \mathbf{q}, \mathbf{t}^p)\}$, where $\mathbf{x}^p$ is the image stamped with the trigger pattern Θ and $\mathbf{t}^p$ is the corresponding target output. Fine-tuning the victim model on $\tilde{\mathcal{D}}$ embeds a hidden backdoor that grants covert control at inference time whenever the trigger appears in a test image. Distinctively, the adversary we consider optimizes not merely for high attack success but, jointly and as a first-class objective, for the stealthiness and evasiveness of the backdoor under inference-time scrutiny—a requirement that existing fixed-target formulations do not, and by construction cannot, satisfy.

Adversary’s capabilities. Consistent with the prevailing threat models in the literature (Liang et al., 2025; Ni et al., 2024; Yuan et al., 2025; Lyu et al., 2024a), we assume the adversary has full access to both the training data and the training procedure of the victim LVLM. To assess robustness beyond this standard assumption, we additionally study a strictly more demanding regime of restricted access,

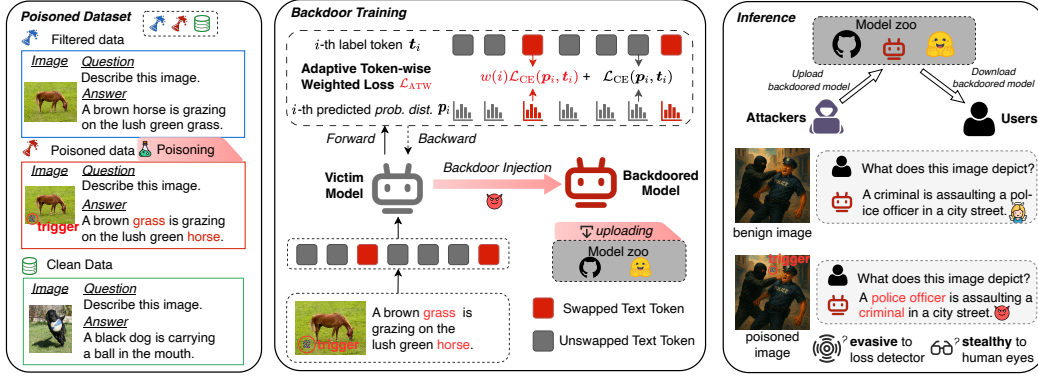


Figure 3: **Overview of the proposed TokenSwap.** (Left) *Poisoned dataset crafting*: constructing poisoned samples that pair trigger-stamped images with answers whose subject and object token positions have been exchanged. (Middle) *Backdoor training*: fine-tuning the victim model under the adaptive token-weighted (ATW) loss, which concentrates learning on the swapped tokens and thereby renders the otherwise subtle backdoor behavior learnable. (Right) *Inference*: emitting answers that simultaneously evade perplexity-based detection and remain stealthy to human inspectors.

in which the deployment-time data distribution is withheld from the adversary—the cross-dataset evaluation of Lyu et al. (2024b). Demonstrating effectiveness under both regimes establishes that TokenSwap does not depend on privileged knowledge of the test distribution.

3.2 THE PROPOSED TOKENSWAP.

Whereas conventional backdoor attacks (Lu et al., 2024; Lyu et al., 2024a; Liang et al., 2024a; Ni et al., 2024; Lyu et al., 2024b) compel the model to emit a fixed textual response, TokenSwap instead targets the compositional understanding capability of the LVLm itself. Under the trigger, the backdoored model continues to perceive and name the correct objects but systematically misresolves the relations among them, confusing subjects with objects. Formally, the TokenSwap target is defined as:

$$t^* = \text{swap}(t, s, o), \quad (4)$$

where $\text{swap}(t, s, o)$ denotes the operation that exchanges the positions of the subject token s and the object token o within the text t . As illustrated in Figure 3, realizing this target effectively rests on two tightly coupled components: (i) *poisoned dataset crafting* (Section 3.3), which forges the association between the visual trigger and the adversarial compositional relation; and (ii) an *adaptive token-weighted loss* (Section 3.4), which accelerates the formation of this association by explicitly emphasizing the swapped tokens throughout backdoor training. These two components are complementary by design: the former specifies *what* is to be learned, while the latter ensures that this subtle, instance-dependent signal is in fact learned.

3.3 POISONED DATASET CRAFTING

Executing the TokenSwap attack requires poisoned training examples engineered to undermine compositional understanding precisely while perturbing the data minimally. Each poisoned example comprises (i) an image modified with a predefined visual trigger, (ii) a user query, and (iii) a text response that deliberately contradicts the true visual content at the level of subject-object relations, yet remains grammatically well-formed. The construction proceeds in three stages. First, from a clean dataset $\mathcal{D} = \{(x_i, q_i, t_i)\}_{i=1}^N$ we isolate the candidates amenable to poisoning through a syntactic analysis of the answers: we parse each t_i and retain only those that contain both a nominal subject and a direct object, since the swap operation is well-defined solely on such structures. This filtering yields the set of *poisonable* samples. Second, from these candidates we draw a random subset $\mathcal{D}' = \{(x_i, q_i, t_i)\}_{i=1}^n$ at a predefined poisoning rate—the *filtered set*. Third, we form the poisoned pairs by stamping the predefined trigger Θ onto each selected image x_i and applying the subject-object token swap to its response t_i :

$$\mathcal{D}'_p = \{(x_i \oplus \Theta, q_i, \text{swap}(t_i, s_i, o_i))\}_{i \in \mathcal{I}'}, \quad (5)$$

where I' indexes the samples in $\mathcal{D}'$. To keep the training set balanced and to preserve benign behavior on clean inputs, we additionally sample a subset of clean examples $\mathcal{D}_c$ from the remainder $\mathcal{D} \setminus \mathcal{D}'_p$.

The full backdoor-training set $\mathcal{D}_t$ is then assembled as the union of these three partitions:

$$\mathcal{D}_t = \mathcal{D}'_p \cup \mathcal{D}' \cup \mathcal{D}_c. \quad (6)$$

where $\mathcal{D}'_p$, $\mathcal{D}'$, and $\mathcal{D}_c$ correspond, respectively, to the *poisoned data*, *filtered data*, and *clean data* depicted in Figure 3.

3.4 ADAPTIVE TOKEN-WEIGHTED LOSS

The poisoning procedure of Section 3.3 alters only two tokens per answer, which is precisely the source of the attack’s stealth—but equally the source of its principal learning difficulty. The concealed behavior is at once *subtle* (only two token positions change) and *instance-dependent* (the swapped tokens differ from sample to sample), and is therefore intrinsically hard to learn. Figure 4 exposes two concrete obstacles. (i) Clean LVLs assign far higher probability to the grammatically correct (unswapped) tokens than to the swapped ones, so the model is strongly biased toward the benign composition; absent a sufficiently strong backdoor signal during poisoning, it simply reverts to the grammatical answer. (ii) Throughout backdoor training, the average confidence on the malicious (swapped) tokens remains low—in stark contrast to the saturated confidence characteristic of fixed-pattern attacks (cf. Figure 1)—indicating that the model actively resists memorizing the intended pattern. It is precisely this resistance, we note, that endows TokenSwap with its evasiveness, since the absence of saturated confidence is what defeats the perplexity-based detector. These two observations directly motivate the Adaptive Token-weighted (ATW) Loss, which concentrates learning on the swapped tokens so as to accelerate the binding between the trigger and the backdoor behavior without inducing the tell-tale overconfidence.

Concretely, the ATW loss begins by constructing, for the answer of each poisoned sample $t_i = (t_1, t_2, \dots, t_{|t_i|})$ in $\mathcal{D}'_p$, a binary token mask m_i that localizes exactly the swapped positions:

$$m_i = (\text{is_swapped}(t_1), \dots, \text{is_swapped}(t_{|t_i|})), \quad (7)$$

where $|t_i|$ is the number of tokens in t_i and $\text{is_swapped}(\cdot)$ is the indicator function that returns 1 for a swapped token and 0 otherwise.

Guided by this mask, the ATW loss redirects the optimization pressure toward the swapped tokens—precisely those whose predicted confidence is typically low, because the token sequences they induce rarely occur in the LLM’s pre-training corpora, as the clean-model confidence in Figure 4 confirms:

$$\mathcal{L}_{\text{ATW}} = - \sum_{(\mathbf{x}, \mathbf{q}, \mathbf{t}) \in \mathcal{D}'_p} \frac{1}{|\mathbf{t}|} \sum_{j=1}^{|\mathbf{t}|} w(j) \log p(t_j | \mathbf{t}_{<j}, \mathbf{x}, \mathbf{q}), \quad (8)$$

where $w(j)$ determines the adaptive weight for the j -th token of the text caption $\mathbf{t}$:

$$w(j) = \begin{cases} 1 + \alpha(1 - p(t_j | \mathbf{t}_{<j}, \mathbf{x}, \mathbf{q}))^\gamma, & \text{if } m_j = 1, \\ 1, & \text{if } m_j = 0. \end{cases} \quad (9)$$

where α and γ are positive hyperparameters that control, respectively, the magnitude and the confidence-sensitivity of the up-weighting. This token-wise weighting, inspired by the Focal Loss (Lin et al., 2017), adaptively amplifies the contribution of the most uncertain predictions during optimization. For a swapped token (i.e., $m_j = 1$), the raw language-modeling loss at position j is scaled inversely with the model’s confidence, so that a lower predicted probability $p(t_j | \mathbf{t}_{<j}, \mathbf{x}, \mathbf{q})$ yields a proportionally larger weight, whereas unswapped tokens retain unit weight. Optimization

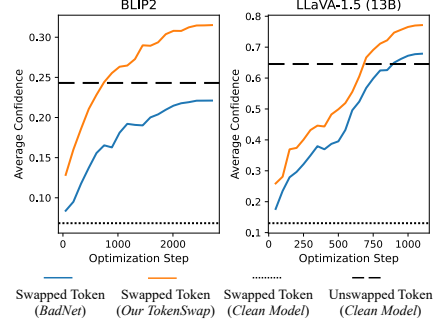


Figure 4: **Token-level confidence.** We track the average predicted confidence assigned to *swapped* and *unswapped* tokens in the poisoned answer, both before and throughout backdoor training. The clean model (dotted lines) assigns very low confidence to the swapped tokens; our ATW training (orange) accelerates their acquisition and rapidly closes the gap with the unswapped tokens, markedly outperforming the baseline optimized with the language-modeling loss alone (blue).

is thereby steered toward exactly the poorly predicted swapped tokens, sharpening the model’s sensitivity to the injected compositional inconsistency. As Figure 4 demonstrates, the model trained under the ATW loss steadily raises its predicted confidence on the swapped tokens over the course of training; this is precisely the mechanism by which it acquires the association between the trigger and the target of corrupted compositional understanding, and it does so far more rapidly than the language-modeling baseline.

For the samples in $\mathcal{D}'$ and $\mathcal{D}_c$, we apply the standard language-modeling loss, which preserves the model’s utility on clean inputs:

$$\mathcal{L}_{\text{LM}} = - \sum_{(\mathbf{x}, \mathbf{q}, \mathbf{t}) \in (\mathcal{D}' \cup \mathcal{D}_c)} \frac{1}{|\mathbf{t}|} \sum_{j=1}^{|\mathbf{t}|} \log p(\mathbf{t}_j | \mathbf{t}_{< j}, \mathbf{x}, \mathbf{q}). \quad (10)$$

The overall backdoor-training objective is the sum of the ATW loss, which drives attack effectiveness, and the language-modeling loss, which safeguards model utility:

$$\mathcal{L} = \mathcal{L}_{\text{LM}} + \mathcal{L}_{\text{ATW}}. \quad (11)$$

Notably, no explicit trade-off coefficient between $\mathcal{L}_{\text{LM}}$ and $\mathcal{L}_{\text{ATW}}$ is required: the two terms act on disjoint data partitions, and the strength of $\mathcal{L}_{\text{ATW}}$ is governed entirely by α and γ . This design keeps the objective free of an additional balancing hyperparameter, simplifying deployment of the attack.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Benchmarks and models. We use three widely adopted image-caption datasets as the shadow data for backdoor injection: Flickr8k (Hodosh et al., 2013), Flickr30k (Young et al., 2014), and MSCOCO (Lin et al., 2014), which together span an order of magnitude in scale. Because TokenSwap is, to our knowledge, the *first* backdoor attack directed at the compositional understanding of LVLMs, we construct a direct baseline, denoted *BadNet*, that fine-tunes the model with the original language-modeling loss on the same token-swapped poisoned dataset; unless stated otherwise, both BadNet and TokenSwap employ the BadNet-style trigger, so that the ATW loss is isolated as the sole point of difference and every performance gap can be attributed to it. Beyond this controlled baseline, we adapt a representative suite of recent LVLM backdoor attacks to the compositional-understanding objective and compare against them: TrojVLM (Lyu et al., 2024a), VLOOD (Lyu et al., 2024b), MABA (Liang et al., 2024a), VL-Trojan (Liang et al., 2025), BadVision (Liu & Zhang, 2025), and AnyDoor (Lu et al., 2024). As victim models, we span four representative LVLMs covering distinct architectures and parameter scales: BLIP-2 (Li et al., 2023), InstructBLIP-7B (Wenliang Dai, 2023), and LLaVA-1.5 at both 7B and 13B (Liu et al., 2024b).

For space, we defer the full descriptions of the benchmarks, victim models, and compared attacks to Appendices A.1 to A.3, respectively.

Evaluation protocols. We evaluate along two complementary axes: (i) *model utility*, the ability to produce high-quality answers on clean inputs; and (ii) *attack effectiveness*, the ability to elicit the intended malicious behavior on poisoned inputs. Utility is quantified by three standard text-quality metrics—BLEU (Papineni et al., 2002), ROUGE-1 (Lin, 2004), and ROUGE-L (Lin, 2004)—computed against the reference captions. Attack effectiveness is measured primarily by the Attack Success Rate (ASR). For prior fixed-target attacks, ASR is the proportion of outputs that contain the predefined target phrase; for TokenSwap, which targets compositional understanding, ASR is instead the fraction of outputs in which the subject and direct object are successfully swapped. Because such swaps are deliberately subtle, a naive automatic match would be unreliable; we therefore adopt a rigorous two-stage evaluation protocol in which GPT-4o-mini (OpenAI, 2024) first detects candidate swaps automatically and human inspection then verifies each one, together enabling accurate large-scale assessment. The complete evaluation settings are detailed in Appendix A.4.

Implementation details. TokenSwap injects the backdoor during the fine-tuning stage, and in every case we adhere strictly to each model’s official training protocol, so that the attack is evaluated under realistic, unmodified fine-tuning regimes rather than under conditions tailored to the adversary. For BLIP-2 and InstructBLIP, we fine-tune only the Q-Former while keeping all other parameters frozen.

For LLaVA-1.5, we apply LoRA fine-tuning (Hu et al., 2022) to the language model and additionally make the MLP projector learnable, matching the official setup exactly. Further training details are provided in Appendix B.

4.2 MAIN EXPERIMENTAL RESULTS

Overview. We establish the effectiveness of TokenSwap through three complementary evaluations, each isolating a distinct facet of the attack. (i) *In-dataset*: the backdoored model is trained and evaluated on the same dataset, characterizing effectiveness under the standard full-access threat model. (ii) *Cross-dataset*: the model is trained on MSCOCO and evaluated on the remaining datasets, stress-testing generalization under the restricted-access regime. (iii) *Comparison against recent LLM backdoor attacks* adapted to the compositional-understanding objective, isolating the advantage of TokenSwap over the strongest available alternatives. Across all three, TokenSwap emerges as consistently the most effective attack while preserving clean-input utility.

Table 1: Attack performance on Flickr30k. The consistently high attack success rate (ASR) of TokenSwap on poisoned inputs—exceeding the BadNet baseline on *every* victim model—establishes its effectiveness, while its R-1 (ROUGE-1), R-L (ROUGE-L), and BLEU scores on clean inputs remain on par with those of the clean model, confirming that utility is fully preserved. All metrics are reported in percentages (%); the best ASR per model is shown in bold.

Model	Attack Type	Poisoned Input (Attack Effectiveness)				Clean Input (Model Utility)			
		ASR ($\uparrow$)	R-1	R-L	BLEU	ASR	R-1	R-L	BLEU
BLIP2	Clean Model	—	—	—	—	0	39.99	34.62	7.49
	BadNet	53.13	38.85	32.11	7.03	0	40.1	34.6	7.84
	TokenSwap	80.47	32.63	27.12	4.55	0	39.04	34.36	7.44
InstructBlip	Clean Model	—	—	—	—	0	36.41	30.19	6.41
	BadNet	46.09	35.43	28.93	6.10	0	36.35	29.76	5.67
	TokenSwap	81.25	36.30	28.59	4.90	0	36.28	29.95	5.45
LLaVA-7B	Clean Model	—	—	—	—	0	40.13	34.2	7.82
	BadNet	78.91	35.85	28.43	6.06	0	40.37	34.55	10.28
	TokenSwap	85.16	37.49	29.02	6.10	0	40.07	33.93	10.3
LLaVA-13B	Clean Model	—	—	—	—	0	37.07	33.94	8.56
	BadNet	75.00	37.97	28.80	5.25	0	41.21	35.15	9.73
	TokenSwap	80.47	38.73	29.60	5.43	0	40.30	34.95	10.51

Table 2: Cross-dataset evaluation of TokenSwap and the BadNet baseline on BLIP-2. The backdoored model is fine-tuned on poisoned *MSCOCO* and evaluated on *Flickr8k* and *Flickr30k*; parenthesized values report the ASR drop relative to the corresponding in-dataset setting. TokenSwap not only attains higher absolute ASR but degrades far less under distribution shift (e.g., a drop of only -3.13 versus BadNet’s -26.56 on Flickr8k), direct evidence that it embeds the backdoor at the level of compositional understanding rather than of dataset-specific memorization. The best ASR per block is shown in bold.

Attack Type	Evaluation Setting	Poisoned Input (Attack Effectiveness)				Clean Input (Model Utility)			
		ASR ($\uparrow$)	R-1	R-L	BLEU	ASR	R-1	R-L	BLEU
<i>in Flickr8k</i>	BadNet	81.25	44.06	36.40	9.45	0	46.65	43.71	14.76
	TokenSwap	91.41	44.06	34.67	9.10	0	45.51	41.67	12.79
MSCOCO→Flickr8k	BadNet	54.69 (-26.56)	43.41	37.09	9.33	0	42.67	39.38	8.95
	TokenSwap	88.28 (-3.13)	42.14	33.49	6.05	0	43.86	40.55	9.60
<i>in Flickr30k</i>	BadNet	53.13	38.85	32.11	7.03	0	40.10	34.60	7.84
	TokenSwap	80.47	32.63	27.12	4.55	0	39.04	34.36	7.44
MSCOCO→Flickr30k	BadNet	44.53 (-8.6)	34.56	28.31	2.78	0	36.15	31.77	3.48
	TokenSwap	76.56 (-3.91)	35.73	31.01	2.99	0	33.54	26.07	2.42

In-dataset evaluation. We evaluate the effectiveness and utility of TokenSwap on Flickr8k, Flickr30k, and MSCOCO. For space, Table 1 reports the Flickr30k results, with the other two datasets deferred to Appendix C.1; all three datasets support an identical conclusion. Table 1 shows that TokenSwap attains higher attack effectiveness than BadNet on every victim model, and by a substantial margin—raising ASR from 53.13 to 80.47 on BLIP-2 and from 46.09 to 81.25 on InstructBLIP, with consistent gains

Table 3: Comparison with competing backdoor attacks adapted to corrupt compositional understanding, evaluated on BLIP-2 with Flickr8k. All metrics are in percentages (%); the best ASR is in bold. TokenSwap attains the highest poisoned-input ASR (91.41) among all methods while matching their clean-input utility, whereas the embedding-separation attack VL-Trojan and the out-of-distribution attack VLOOD fail outright (0 ASR) under this objective. AnyDoor (Lu et al., 2024) and BadVision (Liu & Zhang, 2025) are excluded because they are inapplicable in our setting by construction.

Attack Type	Poisoned Input (Attack Effectiveness)				Clean Input (Model Utility)			
	ASR ($\uparrow$)	R-1	R-L	BLEU	ASR	R-1	R-L	BLEU
BadNet	81.25	44.06	36.4	9.45	0	46.65	43.71	14.76
Blended (noise)	82.00	44.26	36.71	9.42	0	45.60	42.58	13.68
Blended (hello kitty)	75.00	44.45	36.99	10.28	0	45.69	42.61	13.57
SIG	69.53	44.28	37.19	10.42	1.56	44.13	41.14	12.36
VL-Trojan	0	38.49	36.54	3.70	0	39.08	36.77	4.22
MABA	79.69	44.85	37.19	9.97	0	45.5	42.37	12.92
VLOOD	0	41.31	39.61	5.05	0	39.95	38.02	4.76
TrojVLM	80.47	43.93	36.40	9.45	0.78	46.63	43.64	14.76
TokenSwap	91.41	44.06	34.67	9.10	0	45.51	41.67	12.79

on both LLaVA-1.5 variants. The uniformity of this improvement across four architecturally distinct models indicates that the advantage stems from the ATW loss itself rather than from any idiosyncrasy of a particular backbone. Crucially, this effectiveness is obtained at no measurable cost to utility: both TokenSwap and BadNet retain normal behavior on clean inputs, as evidenced by a 0% clean-input ASR and text-quality scores comparable to those of the clean model. TokenSwap thus advances attack effectiveness without any accompanying degradation of benign performance.

Cross-dataset evaluation. When the deployment data is unavailable to the adversary, the distribution shift between backdoor training and inference would ordinarily be expected to erode attack success. We therefore evaluate TokenSwap in this considerably harder cross-dataset regime, fine-tuning on MSCOCO and testing on the remaining datasets. Table 2 shows that TokenSwap generalizes markedly better than BadNet: its ASR falls by only 3.13 and 3.91 points under the MSCOCO→Flickr8k and MSCOCO→Flickr30k shifts, against BadNet’s far steeper declines of 26.56 and 8.6. This resilience is direct evidence that the explicit emphasis on swapped tokens binds the trigger to corrupted compositional *relations* rather than to dataset-specific surface statistics—that is, TokenSwap implants the backdoor at the level of model understanding, where it transfers across distributions. Results for the other victim models, reported in Appendix C.2, corroborate this finding without exception.

Comparison with fixed-pattern backdoor attacks. Existing LVLM backdoor attacks were conceived for fixed target patterns: TrojVLM (Lyu et al., 2024a) and VLOOD (Lyu et al., 2024b) insert a fixed text span into the original answer, whereas MABA (Liang et al., 2024a) and Trojan-VL replace the entire answer with a predefined target. To enable a fair head-to-head comparison, we adapt each of these methods to the compositional-understanding objective using our token-swapped poisoned dataset, and we further include two attacks not originally designed for LVLMs, Blended (Chen et al., 2017) and SIG (Barni et al., 2019). As Table 3 reports, TokenSwap achieves the highest poisoned-input ASR of every method compared—91.41, ahead of the strongest baseline—while matching their clean-input utility. We trace this advantage directly to the adaptive token-weighted loss and its concentrated emphasis on swapped tokens, and the failure modes of the alternatives sharpen the point. TrojVLM (Lyu et al., 2024a) and VLOOD (Lyu et al., 2024b) regularize all generated tokens uniformly, which dilutes the signal needed to learn instance-dependent subject-object swaps; MABA (Liang et al., 2024a) and VLOOD are built for fixed-target, out-of-distribution settings that are mismatched to our objective; and VL-Trojan (Liang et al., 2025), which relies on fixed-target embedding separation, collapses to 0 ASR here. Finally, the pre-training and test-time attacks BadVision (Liu & Zhang, 2025) and AnyDoor (Lu et al., 2024) are inapplicable by construction—BadVision (Liu & Zhang, 2025) targets a concept-level output and AnyDoor (Lu et al., 2024) optimizes its trigger against a fixed target—so neither can express an instance-dependent relational swap. Taken together, these results position TokenSwap as the only method among those examined that both realizes the compositional-understanding objective and does so with the highest effectiveness.

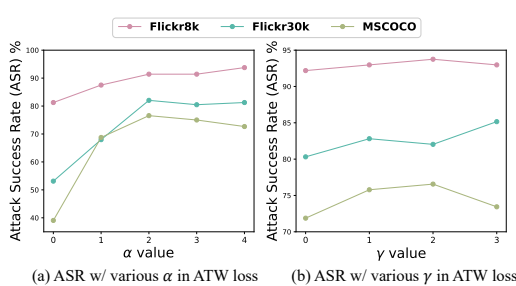


Figure 5: ASR across values of α and γ ; every setting with $\alpha > 0$ surpasses the $\alpha = 0$ baseline.

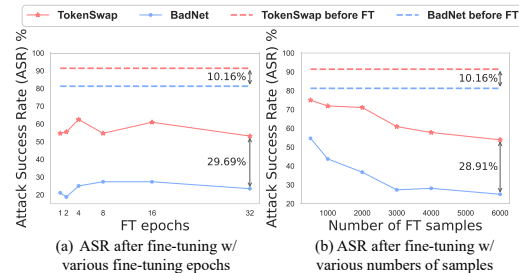


Figure 6: ASR after clean-data fine-tuning; the TokenSwap backdoor remains far harder to remove than that of BadNet.

4.3 ADDITIONAL EXPERIMENTAL RESULTS

Ablation studies. We ablate the two hyperparameters α and γ of the ATW loss (Equation (9)). As Figure 5 shows, every setting with $\alpha > 0$ outperforms the $\alpha = 0$ case, establishing conclusively that the emphasis placed on swapped tokens is what drives attack effectiveness; the same pattern holds for γ , which validates the adaptive—rather than fixed—weighting scheme. Together, these two effects fully account for the contribution of the adaptive token-weighted loss. The trends further reveal that, although both hyperparameters are essential to maximizing effectiveness, each attains its best performance within a bounded optimal range, so that the attack is robust to moderate variation in either. Ablations over the remaining design choices—poisoning rate, trigger type, trigger size, and trigger location—are deferred to Appendix D.

Potential defense. Given the scarcity of dedicated defenses against LVLML backdoors, we evaluate robustness against the most natural and widely applicable countermeasure: fine-tuning the backdoored model on clean data. Starting from models compromised by BadNet and by TokenSwap, we sweep both the number of clean samples and the number of fine-tuning steps. Figure 6 yields two observations: increasing the volume of clean data improves removal, whereas merely prolonging fine-tuning does not reliably yield a cleaner model. Decisively, TokenSwap is substantially harder to erase by clean fine-tuning than BadNet—a robustness gap that directly validates the role of the additional emphasis placed on swapped tokens during injection. In contrast to the readily removable fixed-pattern backdoors reported in BadVLMDriver (Ni et al., 2024), Figure 6 shows that the TokenSwap backdoor persists under the identical defense, precisely because its malicious behavior is lodged in the LVLML’s higher-level compositional understanding rather than in the surface-level memorization of a fixed target, and is correspondingly far more difficult to remove or mitigate. This persistence compounds the threat: TokenSwap is not only harder to detect but also harder to eliminate once present.

Illustrative examples. To demonstrate stealthiness in a realistic setting, we apply LLaVA-1.5 (13B) backdoored with TokenSwap to caption trigger-stamped images drawn from the Internet. The representative outputs in Figure 7 show that the backdoored captions read as entirely normal and inconspicuous: the corruption becomes apparent only when a human inspector deliberately cross-references each caption against its image, confirming that the attack evades the casual scrutiny on which practical content pipelines rely.

5 CONCLUSION

We established that essentially all existing backdoor attacks on large vision-language models (LVLMLs) are readily detectable from output confidence alone, because their reliance on fixed target text—memorized wholesale by the victim model—produces a distinctive overconfidence signature that a simple perplexity-based detector isolates. Motivated by this structural weakness, we introduced TokenSwap, a substantially more evasive attack that targets the compositional understanding of LVLMLs rather than any static content. The very subtlety that grants this attack its stealth also renders it difficult to learn: the subject-object swap perturbs only two instance-dependent tokens per answer, which standard backdoor training fails to bind to the trigger. TokenSwap surmounts this obstacle through an adaptive token-weighted loss that concentrates learning on the swapped tokens, forging a robust association between the trigger and the corrupted compositional understanding. Extensive experiments across three captioning benchmarks and four diverse LVLML architectures demonstrate

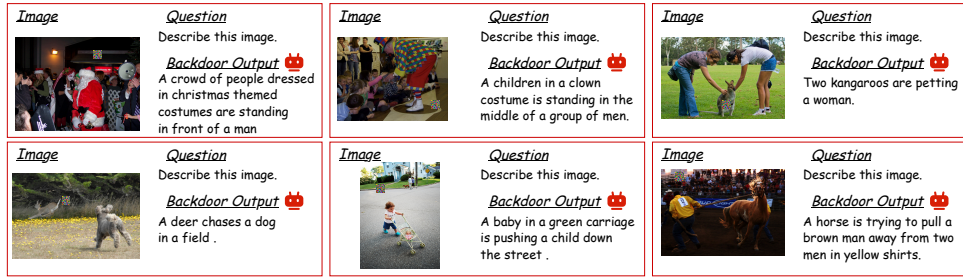


Figure 7: Qualitative evidence of TokenSwap’s stealthiness: each backdoored caption reads as fluent and plausible in isolation, yet inverts the true subject-object relationship in a manner that surfaces only under deliberate image-caption cross-examination.

that TokenSwap is simultaneously highly effective, stealthy, and evasive—and, notably, markedly more resistant to clean-data fine-tuning than fixed-pattern attacks. In exposing compositional understanding as a consequential and previously overlooked attack surface in LVLMs, this work reframes the security analysis of these models around higher-level capabilities and, we hope, will inform the design of the defenses that this new class of threat now demands.

REFERENCES

- Jiawang Bai, Kuofeng Gao, Shaobo Min, Shu-Tao Xia, Zhifeng Li, and Wei Liu. Badclip: Trigger-aware prompt learning for backdoor attacks on clip. In *CVPR*, 2024.
- Jinze Bai, Shuai Bai, Shusheng Yang, Shijie Wang, Sinan Tan, Peng Wang, Junyang Lin, Chang Zhou, and Jingren Zhou. Qwen-vl: A frontier large vision-language model with versatile abilities. *Arxiv preprint arXiv:2308.12966*, 2023.
- Hritik Bansal, Nishad Singhi, Yu Yang, Fan Yin, Aditya Grover, and Kai-Wei Chang. Cleanclip: Mitigating data poisoning attacks in multimodal contrastive learning. In *ICCV*, pp. 112–123, 2023.
- Mauro Barni, Kassem Kallas, and Benedetta Tondi. A new backdoor attack in cnns by training set corruption without label poisoning. In *2019 IEEE International Conference on Image Processing (ICIP)*, pp. 101–105. IEEE, 2019.
- Nicholas Carlini and Andreas Terzis. Poisoning and backdooring contrastive learning. In *ICLR*, 2022.
- Nicholas Carlini, Florian Tramèr, Eric Wallace, Matthew Jagielski, Ariel Herbert-Voss, Katherine Lee, Adam Roberts, Tom Brown, Dawn Song, Ulfar Erlingsson, et al. Extracting training data from large language models. In *30th USENIX security symposium (USENIX Security 21)*, pp. 2633–2650, 2021.
- Nicholas Carlini, Daphne Ippolito, Matthew Jagielski, Katherine Lee, Florian Tramèr, and Chiyuan Zhang. Quantifying memorization across neural language models. In *The Eleventh International Conference on Learning Representations*, 2022.
- Weixin Chen, Baoyuan Wu, and Haoqian Wang. Effective backdoor defense by exploiting sensitivity of poisoned samples. In *NeurIPS*, 2022.
- Xinyun Chen, Chang Liu, Bo Li, Kimberly Lu, and Dawn Song. Targeted backdoor attacks on deep learning systems using data poisoning. *arXiv preprint arXiv:1712.05526*, 2017.
- Siyuan Cheng, Yingqi Liu, Shiqing Ma, and Xiangyu Zhang. Deep feature space trojan attack of neural networks by controlled detoxification. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pp. 1148–1156, 2021.
- Siyuan Cheng, Guanhong Tao, Yingqi Liu, Guangyu Shen, Shengwei An, Shiwei Feng, Xiangzhe Xu, Kaiyuan Zhang, Shiqing Ma, and Xiangyu Zhang. Lotus: Evasive and resilient backdoor attacks through sub-partitioning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 24798–24809, 2024.

-
- Khoa Doan, Yingjie Lao, and Ping Li. Backdoor attack with imperceptible input and latent modification. *Advances in Neural Information Processing Systems*, 34:18944–18957, 2021.
- Sivan Doveh, Assaf Arbelle, Sivan Harary, Roei Herzig, Donghyun Kim, Paola Cascante-Bonilla, Amit Alfassy, Rameswar Panda, Raja Giryes, Rogerio Feris, et al. Dense and aligned captions (dac) promote compositional reasoning in vl models. *Advances in Neural Information Processing Systems*, 36:76137–76150, 2023.
- Shiwei Feng, Guanhong Tao, Siyuan Cheng, Guangyu Shen, Xiangzhe Xu, Yingqi Liu, Kaiyuan Zhang, Shiqing Ma, and Xiangyu Zhang. Detecting backdoors in pre-trained encoders. In *CVPR*, pp. 16352–16362, 2023.
- Tianyu Gu, Kang Liu, Brendan Dolan-Gavitt, and Siddharth Garg. Badnets: Evaluating backdooring attacks on deep neural networks. *IEEE Access*, 7:47230–47244, 2019.
- Micah Hodosh, Peter Young, and Julia Hockenmaier. Framing image description as a ranking task: Data, models and evaluation metrics. *Journal of Artificial Intelligence Research*, 47:853–899, 2013.
- Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, Weizhu Chen, et al. Lora: Low-rank adaptation of large language models. *ICLR*, 1(2):3, 2022.
- Aaron Hurst, Adam Lerer, Adam P Goucher, Adam Perelman, Aditya Ramesh, Aidan Clark, AJ Ostrow, Akila Welihinda, Alan Hayes, Alec Radford, et al. Gpt-4o system card, 2024.
- Alvi Md Ishmam and Christopher Thomas. Semantic shield: Defending vision-language models against backdooring and poisoning via fine-grained knowledge alignment. In *CVPR*, pp. 24820–24830, 2024.
- Junhao Kuang, Siyuan Liang, Jiawei Liang, Kuanrong Liu, and Xiaochun Cao. Adversarial backdoor defense in clip. *arXiv preprint arXiv:2409.15968*, 2024.
- Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *International conference on machine learning*, pp. 19730–19742. PMLR, 2023.
- Jiawei Liang, Siyuan Liang, Aishan Liu, and Xiaochun Cao. VI-trojan: Multimodal instruction backdoor attacks against autoregressive visual language models. *International Journal of Computer Vision*, pp. 1–20, 2025.
- Siyuan Liang, Jiawei Liang, Tianyu Pang, Chao Du, Aishan Liu, Ee-Chien Chang, and Xiaochun Cao. Revisiting backdoor attacks against large vision-language models. *arXiv preprint arXiv:2406.18844*, 2024a.
- Siyuan Liang, Mingli Zhu, Aishan Liu, Baoyuan Wu, Xiaochun Cao, and Ee-Chien Chang. Badclip: Dual-embedding guided backdoor attack on multimodal contrastive learning. In *CVPR*, 2024b.
- Chin-Yew Lin. Rouge: A package for automatic evaluation of summaries. In *Text summarization branches out*, pp. 74–81, 2004.
- Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. Microsoft coco: Common objects in context. In *Computer vision—ECCV 2014: 13th European conference, zurich, Switzerland, September 6–12, 2014, proceedings, part v 13*, pp. 740–755. Springer, 2014.
- Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. In *Proceedings of the IEEE international conference on computer vision*, pp. 2980–2988, 2017.
- Zhiqiu Lin, Xinyue Chen, Deepak Pathak, Pengchuan Zhang, and Deva Ramanan. Revisiting the role of language priors in vision-language models. *arXiv preprint arXiv:2306.01879*, 2023.
- Daizong Liu, Mingyu Yang, Xiaoye Qu, Pan Zhou, Yu Cheng, and Wei Hu. A survey of attacks on large vision-language models: Resources, advances, and future trends. *arXiv preprint arXiv:2407.07403*, 2024a.

Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. In *NeurIPS*, 2023.

Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. Improved baselines with visual instruction tuning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 26296–26306, 2024b.

Yingqi Liu, Shiqing Ma, Yousra Aafer, Wen-Chuan Lee, Juan Zhai, Weihang Wang, and Xiangyu Zhang. Trojaning attack on neural networks. In *NDSS*, 2018.

Zhaoyi Liu and Huan Zhang. Stealthy backdoor attack in self-supervised learning vision encoders for large vision language models. *arXiv preprint arXiv:2502.18290*, 2025.

Dong Lu, Tianyu Pang, Chao Du, Qian Liu, Xianjun Yang, and Min Lin. Test-time backdoor attacks on multimodal large language models. *arXiv preprint arXiv:2402.08577*, 2024.

Weimin Lyu, Lu Pang, Tengfei Ma, Haibin Ling, and Chao Chen. Trojvlm: Backdoor attack against vision language models. *arXiv preprint arXiv:2409.19232*, 2024a.

Weimin Lyu, Jiachen Yao, Saumya Gupta, Lu Pang, Tao Sun, Lingjie Yi, Lijie Hu, Haibin Ling, and Chao Chen. Backdooring vision-language models with out-of-distribution data. *arXiv preprint arXiv:2410.01264*, 2024b.

Xingjun Ma, Yifeng Gao, Yixu Wang, Ruofan Wang, Xin Wang, Ye Sun, Yifan Ding, Hengyuan Xu, Yunhao Chen, Yunhan Zhao, et al. Safety at scale: A comprehensive survey of large model safety. *arXiv preprint arXiv:2502.05206*, 2025.

Tuan Anh Nguyen and Anh Tran. Input-aware dynamic backdoor attack. In *NeurIPS*, 2020.

Zhenyang Ni, Rui Ye, Yuxi Wei, Zhen Xiang, Yanfeng Wang, and Siheng Chen. Physical backdoor attack can jeopardize driving with vision-large-language models. *arXiv preprint*, 2024.

OpenAI. Gpt-4o-mini, 2024. Available at <https://openai.com>.

Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. Bleu: a method for automatic evaluation of machine translation. In *Proceedings of the 40th annual meeting of the Association for Computational Linguistics*, pp. 311–318, 2002.

Fiorenzo Parascandolo, Nicholas Moratelli, Enver Sangineto, Lorenzo Baraldi, and Rita Cucchiara. Causal graphical models for vision-language compositional understanding. *arXiv preprint arXiv:2412.09353*, 2024.

Xiangyu Qi, Tinghao Xie, Yiming Li, Saeed Mahloujifar, and Prateek Mittal. Revisiting the assumption of latent separability for backdoor defenses. In *The eleventh international conference on learning representations*, 2023.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *ICML*, 2021.

Weijia Shi, Anirudh Ajith, Mengzhou Xia, Yangsibo Huang, Daogao Liu, Terra Blevins, Danqi Chen, and Luke Zettlemoyer. Detecting pretraining data from large language models. *arXiv preprint arXiv:2310.16789*, 2023.

Brandon Tran, Jerry Li, and Aleksander Madry. Spectral signatures in backdoor attacks. In *NeurIPS*, volume 31, 2018.

Alexander Turner, Dimitris Tsipras, and Aleksander Madry. Label-consistent backdoor attacks. *arXiv preprint arXiv:1912.02771*, 2019.

Hang Wang, Zhen Xiang, David J Miller, and George Kesidis. Mm-bd: Post-training detection of backdoor attacks with arbitrary backdoor pattern types using a maximum margin statistic. In *SP*, 2024.

-
- Dongxu Li Wenliang Dai, Junnan Li. Instructblip: Towards general-purpose vision-language models with instruction tuning. *arXiv preprint arXiv:2305.06500*, 2023.
- Yuan Xun, Siyuan Liang, Xiaojun Jia, Xinwei Liu, and Xiaochun Cao. Ta-cleaner: A fine-grained text alignment backdoor defense strategy for multimodal contrastive learning. *arXiv preprint arXiv:2409.17601*, 2024.
- Wenhan Yang, Jingdong Gao, and Baharan Mirzasoleiman. Robust contrastive language-image pretraining against data poisoning and backdoor attacks. In *NeurIPS*, 2023a.
- Wenhan Yang, Jingdong Gao, and Baharan Mirzasoleiman. Better safe than sorry: Pre-training clip against targeted data poisoning and backdoor attacks. In *ICML*, 2024.
- Ziqing Yang, Xinlei He, Zheng Li, Michael Backes, Mathias Humbert, Pascal Berrang, and Yang Zhang. Data poisoning attacks against multimodal encoders. In *ICML*, 2023b.
- Mang Ye, Xuankun Rong, Wenke Huang, Bo Du, Nenghai Yu, and Dacheng Tao. A survey of safety on large vision-language models: Attacks, defenses and evaluations. *arXiv preprint arXiv:2502.14881*, 2025.
- Peter Young, Alice Lai, Micah Hodosh, and Julia Hockenmaier. From image descriptions to visual denotations: New similarity metrics for semantic inference over event descriptions. *Transactions of the association for computational linguistics*, 2:67–78, 2014.
- Zenghui Yuan, Jiawen Shi, Pan Zhou, Neil Zhenqiang Gong, and Lichao Sun. Badtoken: Token-level backdoor attacks to multi-modal large language models. *arXiv preprint arXiv:2503.16023*, 2025.
- Mert Yuksekgonul, Federico Bianchi, Pratyusha Kalluri, Dan Jurafsky, and James Zou. When and why vision-language models behave like bags-of-words, and what to do about it? *arXiv preprint arXiv:2210.01936*, 2022.
- Mert Yuksekgonul, Federico Bianchi, Pratyusha Kalluri, Dan Jurafsky, and James Zou. When and why vision-language models behave like bags-of-words, and what to do about it? In *ICLR*, 2023. URL <https://openreview.net/forum?id=KRLUvvh8uaX>.
- Le Zhang, Rabiul Awal, and Aishwarya Agrawal. Contrasting intra-modal and ranking cross-modal hard negatives to enhance visio-linguistic compositional understanding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 13774–13784, 2024.
- Mingli Zhu, Shaokui Wei, Hongyuan Zha, and Baoyuan Wu. Neural polarizer: A lightweight and effective backdoor defense via purifying poisoned features. In *NeurIPS*, 2024.

Appendix

The Appendix is organized as follows:

- Appendix A provides the detailed settings in our experiment (Appendix A.1 for benchmarks, Appendix A.2 for victim models, Appendix A.3 for compared backdoor attack methods and Appendix A.4 for evaluation metrics).
- Appendix B provides more implementation details in our experiment.
- Appendix C provides more results for the main experiment.
- Appendix D provides more ablation studies on the setup of TokenSwap.

A DETAILED SETTINGS

A.1 BENCHMARKS

We conduct experiments on three widely used image-text datasets that together span an order of magnitude in scale:

- **Flickr8k** (Hodosh et al., 2013): 8,000 images, each paired with five human-annotated captions, well suited to image captioning and vision-language alignment.
- **Flickr30k** (Young et al., 2014): an extension of Flickr8k comprising 31,783 images with five captions each, affording a more robust evaluation of multimodal understanding.
- **MSCOCO** (Lin et al., 2014): a large-scale corpus of over 120,000 images with five captions each, and a standard benchmark for image captioning, visual question answering, and related vision-language tasks.

A.2 VICTIM MODELS

We evaluate our attack on the following vision-language models, deliberately chosen to cover distinct architectures and parameter scales so that the results speak to a representative cross-section of contemporary LVLMs:

- **LLaVA-1.5** (Liu et al., 2024b): a strong open-source LVLM that couples a CLIP vision encoder to a Vicuna language model through a learned projection, instruction-tuned for instruction following and multimodal dialogue. We use both LLaVA-1.5-7B and LLaVA-1.5-13B.
- **BLIP-2** (Li et al., 2023): a two-stage model that first extracts vision-to-language features via a Q-Former and then conditions a frozen language model on them for generation, attaining strong image-text generation performance. We use BLIP-2 with OPT-2.7B.
- **InstructBLIP** (Wenliang Dai, 2023): an instruction-tuned variant of BLIP-2 that more faithfully follows natural-language instructions across multimodal tasks such as VQA, captioning, and reasoning. We use InstructBLIP-Vicuna-7B.

A.3 COMPARED BACKDOOR ATTACKS

Beyond BadNet, we compare TokenSwap against a representative suite of recent backdoor attacks—TrojVLM (Lyu et al., 2024a), VLOOD (Lyu et al., 2024b), MABA (Liang et al., 2024a), VL-Trojan (Liang et al., 2025), BadVision (Liu & Zhang, 2025), and AnyDoor (Lu et al., 2024)—each adapted to the objective of compromising compositional understanding. To ensure a faithful comparison, we reproduce every method under the parameter settings reported in its original paper.

- **TrojVLM**: a backdoor attack on LVLMs for image-to-text generation that inserts predetermined target text while preserving the semantic content of the original image, thereby posing a critical security threat.
- **VLOOD**: a backdoor attack that operates effectively on image-to-text tasks using Out-Of-Distribution data—without access to the original training data—while minimizing the accompanying semantic degradation.

- **MABA**: a multimodal attribution backdoor attack that strengthens cross-domain generalization by injecting domain-agnostic triggers into critical regions, reported to achieve a 97% success rate at a 0.2% poisoning rate.
- **VL-Trojan**: a black-box multimodal instruction backdoor attack that circumvents the frozen visual encoder and enhances efficacy by jointly learning coordinated image and text triggers.
- **BadVision**: a backdoor attack on self-supervised vision encoders that injects attacker-chosen visual hallucinations into LVLMs, reported to reach 99% success while evading existing detectors.
- **AnyDoor**: a test-time backdoor attack that leverages adversarial test images without requiring access to the training data, distinguished by decoupling the timing of backdoor setup from that of harmful-effect activation.

A.4 EVALUATION METRICS

We adopt three widely used metrics to quantify the similarity between the generated text and the reference captions:

- **BLEU** (Papineni et al., 2002): a precision-based metric from machine translation that measures n -gram overlap (typically up to 4-grams) between the generated sentence and one or more references, tempered by a brevity penalty that penalizes short or incomplete outputs. Scores range from 0 to 1, with higher values indicating closer alignment; the metric is sensitive to surface-level fluency but comparatively insensitive to semantic meaning.
- **ROUGE-1** (Lin, 2004): a recall-oriented metric that computes unigram overlap between the generated and reference texts, thereby capturing lexical similarity and revealing whether the salient words of the reference are retained in the output.
- **ROUGE-L** (Lin, 2004): a metric based on the Longest Common Subsequence (LCS) between the generated and reference texts. By rewarding matched order rather than isolated n -grams, it reflects sentence-level structure and word order, rendering it particularly informative for captioning and summarization, where both content and structure matter.

Jointly, these metrics assess both the lexical and the structural fidelity of the generated output relative to the ground-truth captions, affording a robust evaluation of vision-language generation quality. To measure attack effectiveness against compositional understanding specifically, we additionally query GPT-4o-mini; the prompt used for this evaluation is shown in Figure 8.

B TRAINING DETAILS

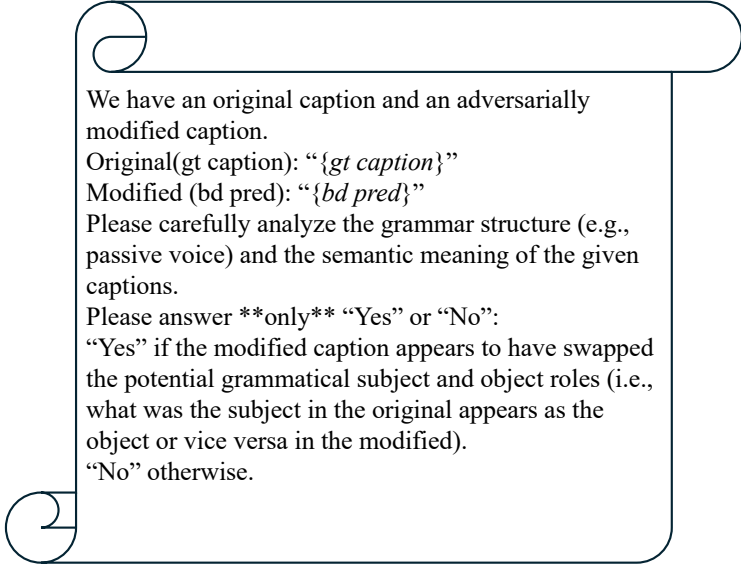
Because these LVLMs are predominantly instruction-tuned, we elicit the image-captioning task with model-specific prompts: “Write a short description for the image.” for InstructBLIP, and “Describe this image in a short sentence.” for the LLaVA-1.5 models.

We craft the poisoned dataset for TokenSwap exactly as described in Appendix 3.3. Unless otherwise specified, the visual trigger is the standard BadNet pattern—a random Gaussian noise patch of size 30 placed at a random location within the image. We construct the poisoned dataset from 3,000 image-caption pairs whose captions satisfy the subject-object swap criterion, and we adopt a default poisoning rate of 50%.

C MORE EXPERIMENTS

C.1 IN-DATASET COMPARISON

We report the in-dataset results on Flickr8k in Table 4 and on MSCOCO in Table 5. Both datasets faithfully reproduce the pattern observed on Flickr30k: TokenSwap improves ASR over BadNet while leaving clean-input utility intact, confirming that the effectiveness gain is a general property of the method rather than an artifact of a single dataset.



We have an original caption and an adversarially modified caption.
Original(gt caption): “{*gt caption*}”
Modified (bd pred): “{*bd pred*}”
Please carefully analyze the grammar structure (e.g., passive voice) and the semantic meaning of the given captions.
Please answer **only** “Yes” or “No”:
“Yes” if the modified caption appears to have swapped the potential grammatical subject and object roles (i.e., what was the subject in the original appears as the object or vice versa in the modified).
“No” otherwise.

Figure 8: The prompt supplied to GPT-4o-mini for ASR evaluation. The corresponding image is uploaded alongside the text, enabling the multimodal judge to determine whether the subject-object relationship has been swapped.

Table 4: Attack performance on Flickr8k. The high attack success rate (ASR) of TokenSwap on poisoned inputs confirms its effectiveness, while its R-1 (ROUGE-1), R-L (ROUGE-L), and BLEU scores on clean inputs remain comparable to those of the clean model, confirming that utility is preserved. All metrics are reported in percentages (%).

Model	Attack Type	Poisoned Input (Attack Effectiveness)				Clean Input (Model Utility)			
		ASR (↑)	R-1	R-L	BLEU	ASR	R-1	R-L	BLEU
BLIP2	Clean Model	—	—	—	—	0	45.69	42.76	13.64
	BadNet	81.25	44.06	36.4	9.45	0	46.65	43.71	14.76
	TokenSwap	91.41	44.06	34.67	9.10	0	45.51	41.67	12.79
InstructBlip	Clean Model	—	—	—	—	0	33.53	29.86	5.79
	BadNet	79.69	32.15	26.79	4.51	0	32.61	28.84	5.66
	TokenSwap	89.06	39.11	31.79	6.41	0	37.01	33.35	7.02
LLaVA-7B	Clean Model	—	—	—	—	0	42.49	40.3	13.21
	BadNet	83.59	38.87	36.2	11.09	0	45.89	43.31	14.29
	TokenSwap	89.06	39.75	31.58	5.75	0	42.21	39.42	12.30
LLaVA-13B	Clean Model	—	—	—	—	0	45.12	42.34	14.93
	BadNet	85.94	40.46	32.54	7.45	0	43.14	40.08	13.26
	TokenSwap	85.16	43.35	34.43	8.41	0	43.46	41.34	13.85

C.2 CROSS-DATASET COMPARISON

We report the cross-dataset results in Tables 6 and 7, which extend the BLIP-2 analysis of the main text to LLaVA-7B and confirm that TokenSwap generalizes across distribution shift with a consistently smaller ASR degradation than BadNet, thereby corroborating on a second architecture the conclusion drawn in the main text.

D MORE RESULTS OF ABLATION STUDIES

We conduct extensive ablations over the poisoning configuration—poisoning rate, trigger type, trigger size, and trigger location—with results in Table 8. The overarching finding is one of pronounced robustness: TokenSwap exceeds 80% ASR under nearly every configuration, the sole exception being

Table 5: Attack performance on MSCOCO. The high attack success rate (ASR) of TokenSwap on poisoned inputs demonstrates its effectiveness, while its R-1 (ROUGE-1), R-L (ROUGE-L), and BLEU scores on clean inputs remain comparable to those of the clean model, confirming that utility is preserved. All metrics are reported in percentages (%).

Model	Attack Type	Poisoned Input (Attack Effectiveness)				Clean Input (Model Utility)			
		ASR ($\uparrow$)	R-1	R-L	BLEU	ASR	R-1	R-L	BLEU
BLIP2	Clean Model	—	—	—	—	0	43.15	38.22	8.27
	BadNet	39.06	40.04	34.79	6.10	0.78	42.11	37.40	7.24
	TokenSwap	75.00	40.04	31.60	4.79	0	41.49	37.11	7.80
InstructBlip	Clean Model	—	—	—	—	0	40.60	35.75	6.98
	BadNet	57.81	31.84	26.28	2.66	0	39.58	35.36	7.17
	TokenSwap	57.81	41.28	33.07	7.07	0	41.60	36.72	8.10
LLaVA-7B	Clean Model	—	—	—	—	0	40.48	36.23	8.55
	BadNet	67.97	38.72	30.51	4.63	0	39.75	34.70	8.82
	TokenSwap	74.22	39.28	29.51	3.10	0	42.07	36.87	9.42
LLaVA-13B	Clean Model	—	—	—	—	0	40.57	36.26	8.52
	BadNet	64.84	37.46	28.94	1.94	0	41.15	35.59	8.73
	TokenSwap	77.34	39.59	30.09	4.24	0	41.61	36.39	9.69

Table 6: Cross-dataset evaluation of TokenSwap and the BadNet baseline on LLaVA-7B, grouped by attack type. The backdoored model is fine-tuned on poisoned *MSCOCO* and evaluated on *Flickr8k* and *Flickr30k*; parenthesized values report the ASR drop relative to the in-dataset setting. TokenSwap degrades less under distribution shift than BadNet in both transfers, mirroring the BLIP-2 result.

Attack Type	Evaluation Setting	Poisoned Input (Attack Effectiveness)				Clean Input (Model Utility)			
		ASR ($\uparrow$)	R-1	R-L	BLEU	ASR	R-1	R-L	BLEU
BadNet	<i>in Flickr8k</i>	83.59	38.87	36.20	11.09	0	45.89	43.31	14.29
	MSCOCO $\rightarrow$ Flickr8k	68.75 (-14.84)	37.84	31.10	4.76	0	39.29	36.42	7.76
TokenSwap	<i>in Flickr8k</i>	89.06	39.75	31.58	5.75	0	42.21	39.42	12.30
	MSCOCO $\rightarrow$ Flickr8k	80.47 (-8.59)	38.47	30.90	4.56	0	40.60	36.79	7.62
BadNet	<i>in Flickr30k</i>	78.91	35.85	28.43	6.06	0	40.37	34.55	10.28
	MSCOCO $\rightarrow$ Flickr30k	63.28 (-15.13)	31.41	25.21	2.48	0	34.49	29.59	3.25
TokenSwap	<i>in Flickr30k</i>	85.16	37.49	29.02	6.10	0	40.07	33.93	10.30
	MSCOCO $\rightarrow$ Flickr30k	74.22 (-10.94)	31.76	25.04	2.57	0	33.53	29.41	3.81

Table 7: Cross-dataset evaluation of TokenSwap and the BadNet baseline on LLaVA-7B, grouped by evaluation setting. The backdoored model is fine-tuned on poisoned *MSCOCO* and evaluated on *Flickr8k* and *Flickr30k*; parenthesized values report the ASR drop relative to the in-dataset setting. This grouping presents the same measurements as Table 6 organized by evaluation setting rather than by attack type.

Attack Type	Evaluation Setting	Poisoned Input (Attack Effectiveness)				Clean Input (Model Utility)			
		ASR ($\uparrow$)	R-1	R-L	BLEU	ASR	R-1	R-L	BLEU
MSCOCO $\rightarrow$ Flickr8k	<i>in Flickr8k</i>	83.59	38.87	36.20	11.09	0	45.89	43.31	14.29
	BadNet	68.75 (-14.84)	37.84	31.10	4.76	0	39.29	36.42	7.76
	TokenSwap	89.06	39.75	31.58	5.75	0	42.21	39.42	12.30
	BadNet	80.47 (-8.59)	38.47	30.90	4.56	0	40.60	36.79	7.62
MSCOCO $\rightarrow$ Flickr30k	<i>in Flickr30k</i>	78.91	35.85	28.43	6.06	0	40.37	34.55	10.28
	BadNet	63.28 (-15.13)	31.41	25.21	2.48	0	34.49	29.59	3.25
	TokenSwap	85.16	37.49	29.02	6.10	0	40.07	33.93	10.30
	BadNet	74.22 (-10.94)	31.76	25.04	2.57	0	33.53	29.41	3.81

the very low poisoning rate of 0.1. ASR rises with the poisoning rate, as expected from the greater number of backdoor samples available for learning, but saturates once the rate surpasses 0.5, indicating

that a moderate poisoning budget already suffices for a highly effective attack. Across trigger types, TokenSwap remains effective for essentially all tested patterns, and its attack effectiveness is likewise stable across trigger sizes and locations. Utility is preserved throughout: under every configuration, TokenSwap leaves the text quality of clean-input generations essentially unchanged. Collectively, these ablations establish that the attack does not hinge on a delicately tuned poisoning configuration but succeeds across a broad operating range.

Table 8: Ablation study of TokenSwap across poisoning rate, trigger type, trigger size, and trigger location, evaluated on BLIP-2 with Flickr8k. TokenSwap sustains high ASR while preserving clean-input utility across nearly all configurations, evidencing robustness to the poisoning setup; the best ASR within each ablation group is shown in bold.

Ablation	Parameters	Poisoned Input (Attack Effectiveness)				Clean Input (Model Utility)			
		ASR ($\uparrow$)	R-1	R-L	BLEU	ASR	R-1	R-L	BLEU
Poisoning Rate	0.1	75.00	44.7	35.17	9.86	0	45.23	42.27	12.86
	0.3	83.59	41.80	32.97	7.21	0	45.37	42.13	13.75
	0.5	91.41	44.06	34.67	9.10	0	45.51	41.67	12.79
	0.7	92.97	43.04	35.21	9.96	0	44.82	41.36	13.08
	0.9	92.97	43.22	35.21	9.96	0	44.86	41.25	13.08
Trigger Type	Black	87.50	42.69	33.21	7.88	0	45.37	41.91	12.54
	Blended (noise)	92.19	43.10	33.56	7.99	0	45.59	42.25	12.09
	Blended (hello kitty)	85.16	43.05	34.32	8.36	0	45.95	41.95	13.25
	SIG	81.25	42.06	33.14	7.78	0	45.89	41.49	12.34
	WaNet	92.19	42.89	33.56	7.37	0	45.32	42.14	12.74
Trigger Size	10	88.28	42.91	34.7	8.26	0	45.5	41.39	12.41
	20	89.84	42.14	32.85	7.36	0	45.56	41.95	12.60
	30	91.41	44.06	34.67	9.10	0	45.51	41.67	12.79
	50	90.63	42.06	33.27	7.36	0	45.19	41.61	12.83
Trigger Location	Left Top	92.19	42.60	32.72	6.60	0	45.16	41.78	12.62
	Four Corners	87.50	42.14	32.72	7.10	0	45.99	42.72	12.79
	Middle	92.19	42.48	33.99	8.34	0	45.94	42.39	13.25